Chapter 2

The LHC and CMS Detector

2.1 The Large Hadron Collider

The Large Hadron Collider, which replaced the prexisting Large Electron-Positron Collider, was constructed between 2000 and 2008. Designed to reach new energy fronteirs by colliding protons at up to 14 TeV, the LHC provided particles physicists with their most powerful tool yet to test the Standard Model and beyond. At the most basic level, these proton-proton collisions are achieved by crossing two counter-rotating proton beams at specific points. These counter-rotating beams are accelerated along the 26.7 km circumference of the LHC within a tunnel of 3.7 m internal diameter [13]. Initial designes proposed two completely separated proton beam pipes, but it became clear that the space requirements of the existing CERN tunnel (constructed for LEP) precluded this possibility. As a result, a ”two-in-one” superconducting magnet design was adopted that allowed the two beams to be transported in separate magnetic coils sharing the same mechanical and cryostat space.

The details of the physical construction of the LHC will be given below in section 2.1.1. Of greater importance, however, are the details of how the proton beams themselves collide, and the various measurements and useful quantities defined therein. In section 2.1.2, many details of the proton-proton will be expounded. This section will define several vital concepts that will be of great interest for the future secions of this paper.

2.1.1 LHC Design

The following section is summarized from [13]. A simplified schematic of the LHC experiment can be seen in figure 2.1. As described above, the two proton beams are designed to circulate in opposite directions and interract at specific points along the beams. These two beams can be seen in red (clockwise) and blue (counterclockwise). Of note, there are 4 interaction points, each of which is surrounded by a specific experiment. The ALICE experiment is designed to measure ion collisions. The LHC-B experiment is specifically designed to measured B-physics. The ATLAS and CMS detectors, on the other hand, are designed as complementary general purpose detectors aimed at leveraging the very high production rate of the LHC. While there are major differences in the designs of the two detectors, they were in essence created to attempt to measure similar physical processes. A major advantage to this is reproducability; whenever CMS or ATLAS records a novel result, the other detector can be used to confirm it.

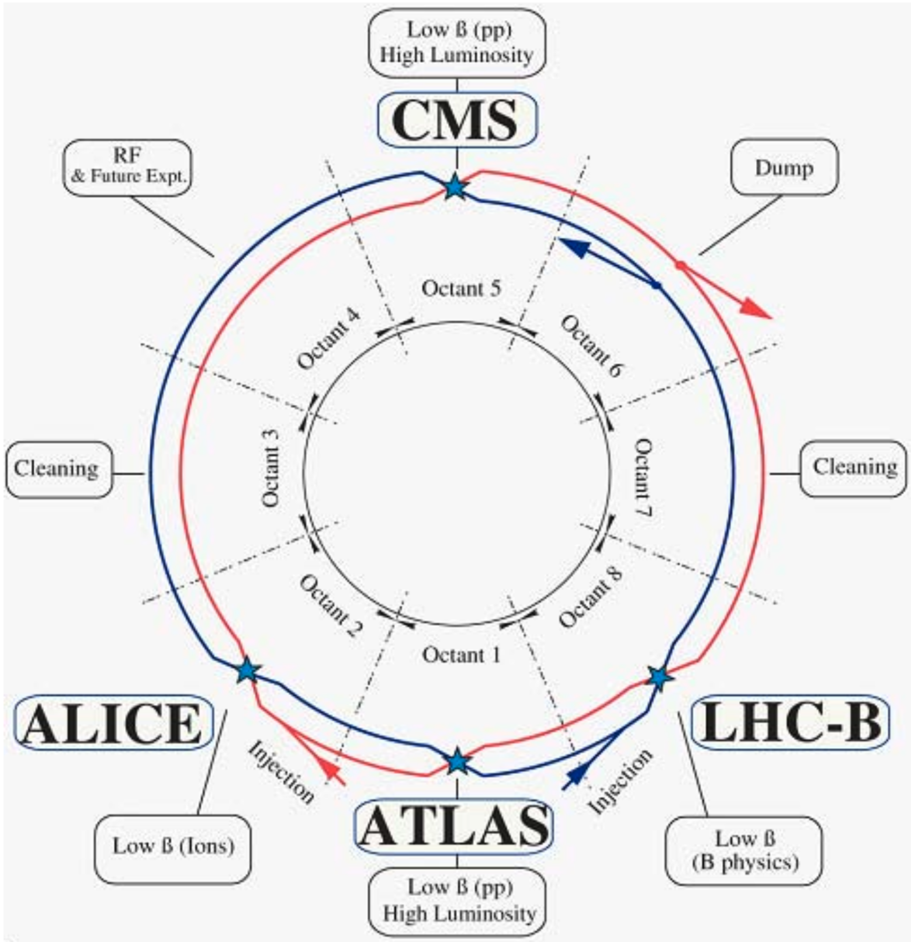

Figure 2.1. A simplified schematic of the overall layout of the LHC. The counter-rotating beams are shown in red (clockwise) and blue (counterclockwise). Reproduced from [13].

2.1.2 Collisions at the LHC

At each of the interaction points shown in figure 2.1, the proton bunches cross over and the individual protons within each bunch interact with each other. There are two major quantities that define the character of these interactions: center of mass energy and luminosity. Center of mass energy, $\sqrt{s}$, is defined as the total combined energy of both protons in their center of mass frame of refernce (where the two protons are moving in opposite directions at equal speeds). This directly measures how much energy is available to produce additional particles. The original design of the LHC allowed collision energies up to 14TeV, though Run3 at the LHC has only been able to achieve a value of $\sqrt{s} = 13.6\text{TeV}$ [1].

The luminosity is a quantity of major concern for physics at the LHC, as it is a measure of the total number of possible collisions in a collider experiment. A higher luminosity value corresponds to more collisions, which corresponds to a higher rate of particle production. The luminosity is measured in two ways. Firstly, the instantaneous luminosity, given in units of $cm^{-2}s^{-1}$, measures the theoretical number of collisions occuring at any given moment. It is defined based on the beam parameters, as shown in equation 2.1.

$$\mathscr{L}_{inst} = \frac{N_b^2 n_b f_{rev} \gamma_r}{4\pi \varepsilon_n \beta^*} F \tag{2.1}$$

Here, N_b is the total number of particles per bunch, n_b is the number of bunches per beam, f_{rev} is the revolution frequency, γ is the standard relativistic factor, ε_n is the normalized transverse beam emmittance, β^* is the beta function at the collision point, and F is the geometric luminosity reduction factor based on the crossing angle of the two beams [13]. The $\mathscr{L}_{inst}$ is a very useful quantity to understand collisions at the LHC, as it also provides an understanding of the derived quantity termed pileup (PU). Pileup measures the number of total collisions that happen per bunch crossing; when attempting to measure a single signal, a higher PU value will increase the amount of background and extraneous signal events occuring simultaneously. For Run 3, the PU averages about 57 collisions per bunch crossing (REF? LHCP Conference Paper).

Beyond the instantaneous luminosity, a major measure used for considering LHC data is the integrated luminosity, $\mathscr{L}_{int}$ (usually measured in inverse femtobarns, fb^{-1}). This is obtained by integrating the instantaneous luminosity over a specific run period. This integration is not as direct as it may seem, however. In reality, the instantaneous luminosity over a specific run decays over time, with a characteristic decay time. This total decay time, τ_L depends on the losses of protons within the bunch due to collisions, the scattering of particles on residual gasses in the beamline, and IBS scattering effects [13]. In terms of this characteristic decay time, the total integrated luminosity can be given by equation 2.2.

$$\mathscr{L}_{int} = L_0 \tau_L \left[1 - e^{-T_{run}/\tau_L}\right] \tag{2.2}$$

Where L_0 is the initial instantaneous luminosity of the run, and T_{run} is the total length of the run. In practice, these quantities are determined by the luminosity group and publicly published for each data sample. Monte Carlo samples are generated with an associated effective luminosity, $\mathscr{L}_{eff}$, which corresponds to the amount of integrated luminosity in the data that would be required to obtain the generated number of signal events.

For any given signal process, as described in chapter 1, there is a cross section, σ, which helps calculate the expected number of signal events for that process. These σ values are measured in some fraction of a barn, which has units of $10^{-28}m^2$. Generally, cross sections are measured in picobarns (pb) or femtobarns (fb). Immediately, it is obvious that these units are the inverse of the total integrated luminosity. So, then, the expected number of total signal events within a given data period with $\mathscr{L}_{int}$ can be given by equation 2.3.

$$N_{event}^{Tot.} = \mathscr{L}_{int}\sigma_{event} \tag{2.3}$$

2.2 The Compact Muon Solenoid Detector

This section is largely reproduced from [11] and [4]. The Compact Muon Solenoid is designed as a 22 m cylinder with a total diameter of 15 m. Centered on the beampipe where a proton-proton bunch crossing occurs, the detector consists of various layers of detector elements designed to track and measure different resultant particles. The general structure of the detector will be described beginning at the center and moving outward. A schematic of the detector can be seen in Figure 2.2. Later sections will provide additional detail on elements of particular interest for our $H \to \gamma\gamma$ analyses.

After particles collide and are scattered, they first interact with the central beampipe

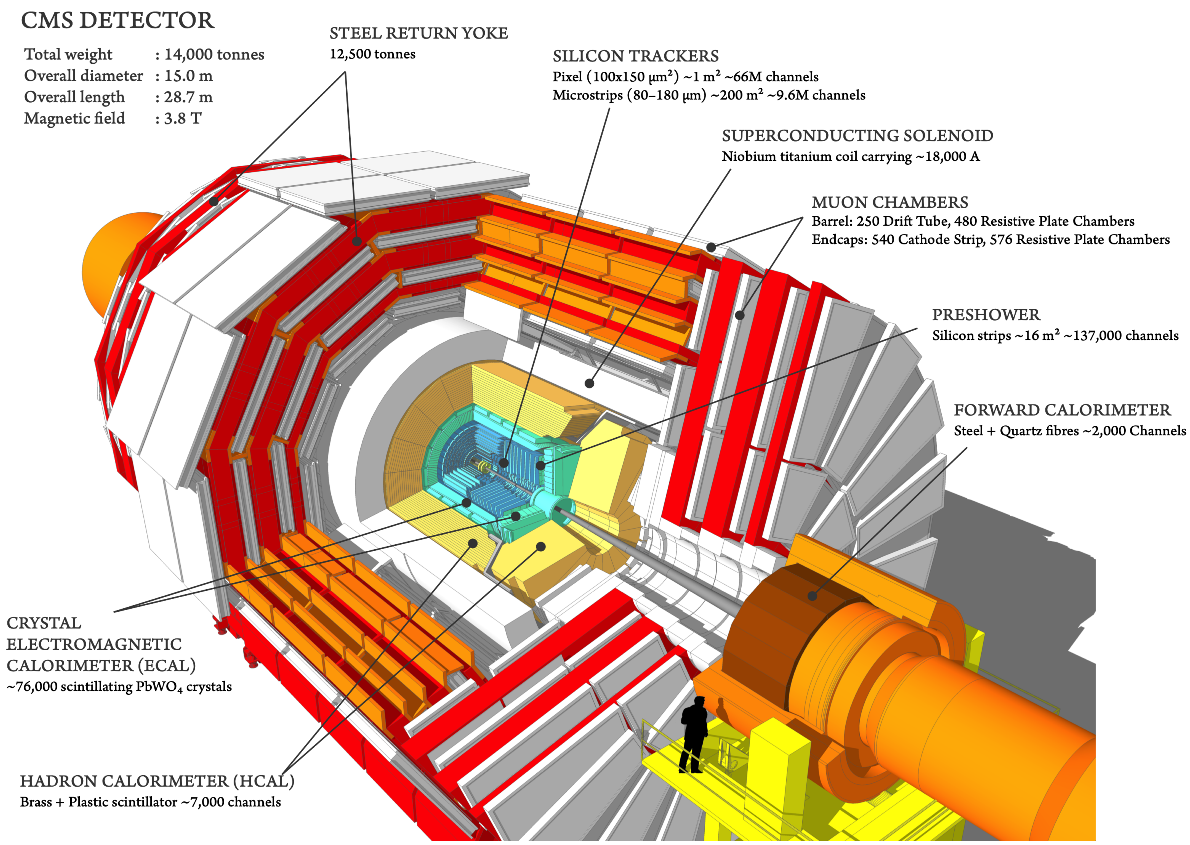

Figure 2.2. A full view of the CMS Detector, with each major system labled. Reproduced from [4].

that houses the incoming proton bunches. While this element does not provide any detecting capabilities, interactions between the outgoing particles and the material of the beampipe may cause additional scattering and particle production. Next, the particles enter the robust inner tracking system, designed to be able to reconstruct the path of charged particles through the detector. This inner tracking system is composed of an internal pixel detector surrounded by an externam strip detector. These elements will be described in section 2.2.1.

The next detector element encountered is the Electromagnetic Calorimeter (ECal), which aims to measure the energies and locations of electrons and photons. Of course, given that the following analysis will be centered on photons, this detector element will be of great interest. This ECal is composed of scintillating $PbWO_4$ crystals arranged into both a cylindrical ECal Barrel (EB) concentric with the beamline and two circular ECal Endcaps (EE) oriented orthogonally to the beamline. A Preshower detector is inserted in front of the EE detectors. More details on this detector element will be provided in section 2.2.2.

Further from the beamline, the Hadronic calorimeter (HCal) aims to measure the locations and energies of charged and neutral hadronic particles using plastic scintillator tiles imbeded in a brass volume. Similarly to the ECal, the HCal is composed of a barrel (HB) region and two endcaps (EB). This detector will be described in more detail in section 2.2.3.

A superconducting solenoid encases all three of these internal regions, providing an internal magnetic field of 3.8T. This magnetic field allows the paths of charged particles to be bent, providing more precise measurement of their properties. While vital to the physics of the detector, this solenoid does not contain any detecting elements. Beyond this magnet lies the highly complicated Muon chambers, which will not be described for this anlysis. Close to the beamline but at the ends of the z-axis of the detector lies the forward calorimeter, which will also not be described for this analysis. The entire detector is shrouded in a steel return yoke.

2.2.1 Beampipe and Inner Tracking System

A major goal of the CMS detector is to reconstruct the tracks of charged particles. This allows the software to not only track and measure the flight of an individual particle, but it allows the primary interaction vertex (PV) to be measured more accuretely. This is achieved by a two-layerd approach; the pixel detectors closest to the beamline prived high-accuracy three-dimmensiional spatial information on the particles immediately after they leave the beampipe, while the strip detectors are used to reconstruct where the particle went after this precise initial location. Since the analysis of $H \to \gamma\gamma$ is focused on photons, charged tracks are not expected for the final state particles of interest. This does not mean they are of no use to us, however – charged tracks can be used to exclude particles from out final state analysis, allowing us to reduce the number of resultant particles and increase the likelihood that the final particles analyzed are photons.

Pixel Detector

The pixel detectors achieve high spatial resolution by using 124 million channels each composed of a single pixel. As with most detector elements at CMS, the pixel detectors are formed by a barrel region (BPIX) and two circular disk endcap regions (FPIX). The barrels use 4 layers of silicon pixels, located at 29, 68, 109, and 160 mm from the beamline. The endcap detectors use 3 layers of pixels, at distances of 291, 396, and 516 mm from the center of the detector along the z-axis.This layout ensures that any charged particle will strike the pixel trackers 4 times at varying distances from the interaction vertex, allowing precise reconstruction of the track starting at the edge of the beampipe.

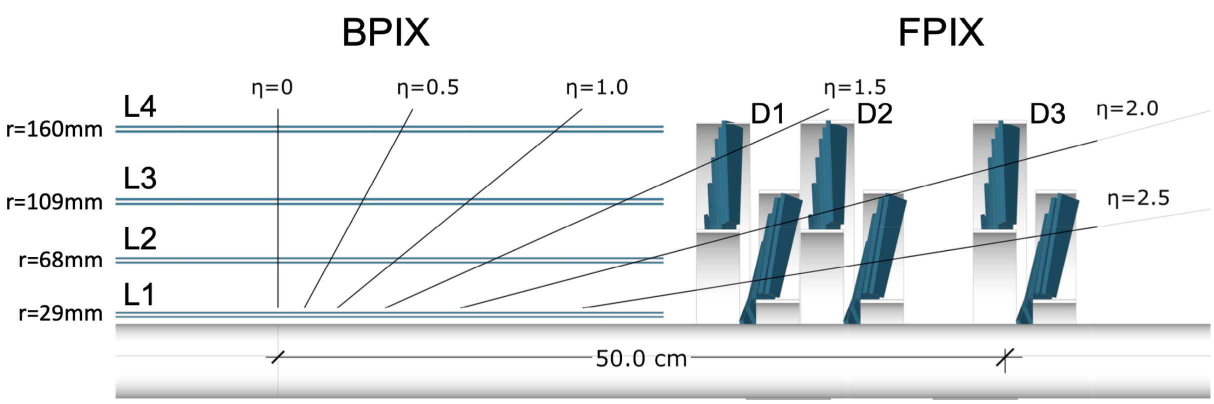

Figure 2.3. Longitudinal view of the pixel detector, where the beampipe is represented at the bottom of the image. Reproduced from [4]

The individual pixels, of the same size in both the BPIX and FPIX, are housed on sensor modules with 160x416 pixels each (with a total size of 100 x 150 μm^2). This modular structure allows easy replacement of faulty modules, as well as allowing more efficienct production. The BPIX contains 1184 such modules, across its 4 layers, whereas the FPIX contains 672 modules. Light-weight mechanical structures provide the mounting points for these modules, and include cooling aparati.

Strip Detector

The innermost pixel detector is rather small, only extending 160mm radially from the beamline, and 516mm along the beam direction from the interaction point. The majority of the tracking system is composed of the silicon strip tracker (SST). The SST contains 9.3 million silicon micro-strips grouped into 15,148 modules. While there are far fewer channels in the SST, the volume covered is much larger; once the origin of a particle track is pinpointed by the pixel detectors, the strip detectors are able to detect the particle as it travels through a large volume of the detector. The SST has a diameter of 2.5m and a length along the beamline of 5m, so it is clearly a more coarse-grain tracking sensor.

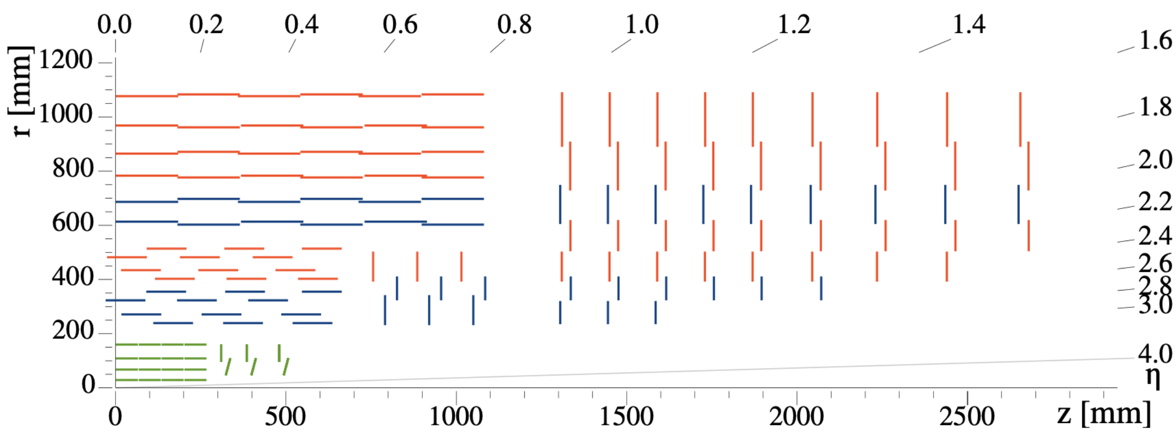

Figure 2.4. Full view of the tracker system. The inner pixel tracker is shown in green. Reproduced from [4]

2.2.2 Electromagnetic Calorimeter

Beyond the inner tracking system, the ECAL system is designed to measure the energies, locations, and arrival times of electrons and photons. A schematic of this detector element can be seen in figure 2.5, reproduced from [11]. Given that this paper will be dealing with a two-photon final state, the details of this detector are highly relevant. Similarly to the inner tracker, the ECAL is composed of modules of individual detector elements. Here, however, since the goal is to measure photons and electrons, scintillating material is required. As a result, the individual detector elements are composed of scintillating lead tungstate ($PbWO_4$) crystals. These crystals were chosen because, in addition to being optically clear, they are radiaton-hard. Given that the detector will be exposed to high levels of radiation for long continuous stretches, it is imperative that the crystals need not be replaced frequently. The photons from the scintillation of these crystals are detected by photodetectors. The photodetectors differ, however, between the barrel and endcap as explained in detail below.

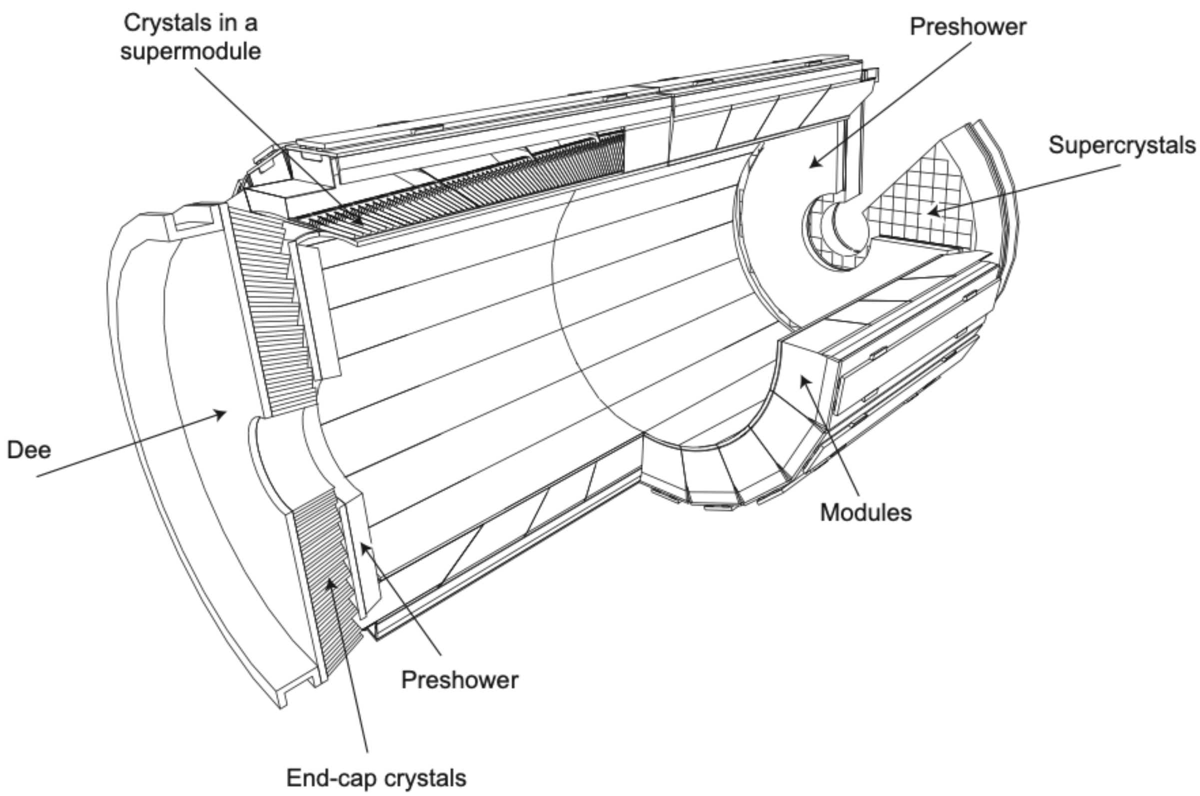

Figure 2.5. A simplified schematic of the ECal. Reproduced from [11]

ECal Barrel (EB)

The barrel component of the ECAL (EB) is designed, of course, with cylindrical symmetry. In the azimuthal direction, ϕ, there are 360 crystals per ”ring”. The barrel detector coveres the pseudorapidity range of $|\eta| < 1.479$, and there are 180 crystals per ”row” in this range. As a result, the EB consists of 61,200 total $PbWO_4$ crystals. These crystals are tapered, such that the scintillated photons are funneled to the back of the crystal where two avalanche photodiodes (APDs) per crystal collect them. The cross-sectional area of the crystal faces is 22x22 mm^2. The face of each crystal lies at a distance of 1.29m from the center of the beamline.

As with the tracker, the individual crystals are combined into submodules and modules, where readout and power-supplying electronics are consolidated. These submodules are of an ”aveolar” structure, where the dividing walls between individual crystals ar 0.1mm thick and are made of a combination of alluminum and fibre-epoxy resin. This leads to a total crystal-to-crystal distance of 0.35mm within each submodule. This provides sufficient granularity while still ensuring complete geometric coverage with a reasonable number of individual channels.

ECal Endcaps (EE)

The ECal endcaps (EE) cover the pseudorapidity range of $1.479 < |\eta| < 3.0$, and are located symmetrically 315.4cm from the interaction point. The EE is designed as a symmetric disk, with identically shaped crystals grouped in units of 5x5 crystals. The crystals are pointed at a focus point 1300mm beyond the interaction point, meaning that they are angled towards the beam line with an increasing angle (between 2 and 8 degrees) as they move further from the center of the endcap. The cross-sections of these crystals are slightly larger than those in the barrel, measuring at 28.62x28.62 mm^2. In total, there are 7,324 crystals for each of the two endcaps laid out in a regular x-y grid. Since this detector element represents much less coverage in terms of area, there are of course fewer total crystals. The difference in the crystal sizes and arrangements relative to the EB means that many of the geometric quanities measured by later reconstructions will differ in the EE.

2.2.3 Hadron Calorimeter

While the hadron calorimeter can provide useful measurements for rejecting hadronic backgrounds from the desired photon signatures of this analysis, most measurements it provides are not of great use for the $H \to \gamma\gamma$ process. As a result, only a basic understanding of this detector will be summarized from [11]. As with the ECal system, the HCal consists of a barrel (HB) and endcap (HE) detector element. Located further from the interaction point than the ECal, these elements are much larger than the more inner detector elements. Still, the major operating principle is to cause scintillation and therefore measure energy deposition in individual crystals. As with the ECal, this allows the examination of the energy deposition in both size and location.

The HB consists of 36 identical "wedges" arranged azimuthally around the beamline, and extend between the end of the EB (at R = 1.77m) and the start of the magnetic coil (R - 2.95m). As a result, these HB wedges are considerably thicker than the ECal detector elements. The HB extends less in η than the EB, covering the area between $|\eta| < 1.3$. Since the barrel extends away from the interaction point (in z), the interaction length, λ_I increases with increasing distance from the interaction point. As a result, at higher η values, signatures pass through considerably more of the detector. The HE consists of two circular caps that cover the range $1.3 < |\eta| < 3.0$.

2.2.4 The Muon Detector

The final, outermost detector element is the muon detector. This is the crowning jewel of the CMS detector, and was a major motivator for its construction. The muon system is designed entirely to measure muon signatures across the entire kinematic range of the LHC. As with the two calorimeter elements, the muon system is designed as a large barrel detector with two additional endcaps covering the higher-η regions near the beamline. While this impressive system marks a monumental step towards precision measurements of processes containing muons (such as Z boson decay), it has little relation to the measurement of $H \to \gamma\gamma$ signatures. The general size and shape of the muon system can be seen in figure 2.2, but no further details need be supplied for this analysis.